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1.1 Characteristics & Variety of Living Organisms

Edexcel International GCSE Biology (4BI1) — Paper 1B/1BR

1.1.1 The Characteristics of Living Organisms

All living organisms share **eight characteristic life processes**, remembered by the mnemonic **MRS GREN C**. These processes distinguish living things from non-living matter and are the foundation of biology.

Letter	Process	What it means	Example
M	Movement	An action by an organism or part of an organism that changes position or place.	A cheetah running; a sunflower head tracking the sun.
R	Respiration	Chemical reactions in cells that break down nutrient molecules to release energy (ATP).	Glucose + oxygen → CO_2 + water in mitochondria of all cells.
S	Sensitivity	The ability to detect and respond to changes in the internal or external environment (stimuli).	Pupil reflex in bright light; roots growing towards water.
G	Growth	A permanent increase in size and dry mass through increase in cell number or size.	A tree growing taller; a child increasing in height and mass.

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R	Reproduction	Processes that make more of the same kind of organism. Can be sexual or asexual.	Humans producing children; strawberry runners forming new plants.
E	Excretion	Removal of toxic waste products of metabolism from the body.	Exhaling CO_2; excreting urea in urine.
N	Nutrition	Taking in and using nutrients (organic substances and minerals) to provide energy, growth and maintenance.	Plants — photosynthesis. Animals — ingestion and digestion of food.
C	Control (homeostasis)	Maintenance of a stable internal environment in response to change.	Regulation of body temperature, blood glucose and water balance in humans.

Exam Tip: Edexcel will often ask "State the eight characteristics of living organisms" for 2–4 marks. **Learn MRS GREN** by rote — it never changes and is guaranteed free marks. Be careful: "photosynthesis" is NOT one of the eight (it is included under Nutrition).

Distinguishing life processes

- **Excretion vs egestion**: Excretion is the removal of metabolic waste (e.g. CO_2 from respiration, urea from deamination). Egestion is the removal of undigested food — not produced by metabolism — and is NOT one of the eight.
- **Respiration vs breathing**: Respiration is the chemical process in cells that releases energy. Breathing (ventilation) is the physical movement of air in and out of the lungs to supply oxygen for respiration.
- **Growth vs increase in mass**: A balloon inflated with gas increases in mass but does not grow. True growth is a permanent increase in dry mass due to new cells or new biomass.

1.1.2 The Five Kingdoms of Living Organisms

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All living organisms are classified into **five kingdoms**, grouped by cell type and by how they obtain nutrients. Three features matter for classification: whether cells are **prokaryotic or eukaryotic**, whether organisms are **unicellular or multicellular**, and their **mode of nutrition**.

The Five Kingdoms

Animalia	Plantae	Fungi	Protoctista	Prokaryota
Eukaryotic Multicellular No cell wall No chloroplasts Heterotrophic (ingest food)	Eukaryotic Multicellular Cellulose cell wall Chloroplasts Autotrophic (photosynthesis)	Eukaryotic Mostly multicellular Chitin cell wall No chloroplasts Saprotrophic (extracellular digestion)	Eukaryotic Usually unicellular Very diverse! Some photosynthesise Others are heterotrophic	Prokaryotic (no true nucleus) Unicellular Peptidoglycan wall DNA as a loop + plasmids
Examples: Human, bird, fish, insect, earthworm	*Examples:* Flowering plants, ferns, mosses, conifers	*Examples:* Mucor, Penicillium, yeast (unicellular), mushrooms	*Examples:* Amoeba, Chlorella, Plasmodium, Spirogyra	*Examples:* Lactobacillus, Pneumococcus, E. coli

All five kingdoms contain eukaryotes — except Prokaryota, which is the only kingdom of prokaryotic organisms.
Viruses are not placed in any kingdom — they are not considered to be truly alive.

The Animalia kingdom

- Eukaryotic multicellular organisms with no cell wall and no chloroplasts.
- Heterotrophic — they consume complex organic molecules made by other organisms (ingestion, then internal digestion).
- Usually show nervous coordination and are motile at some stage of their life.
- Store carbohydrate as glycogen in muscles and liver.

The Plantae kingdom

- Eukaryotic multicellular organisms with a **cellulose cell wall**, **chloroplasts** containing chlorophyll, and a **large permanent vacuole**.
- Autotrophic — make their own food by photosynthesis.
- Store carbohydrate as starch; use sucrose for transport in phloem.
- Examples include flowering plants (Magnoliophyta), ferns, mosses, and conifers.

The Fungi kingdom

- Eukaryotic, mostly multicellular organisms. Bodies made of thread-like **hyphae** that form a network called the **mycelium**. (Yeast is an important unicellular exception.)
- Cell walls of **chitin** (not cellulose).
- Heterotrophic and feed by **saprotrophic nutrition** — secrete digestive enzymes onto dead organic matter and absorb the soluble products (extracellular digestion).
- Reproduce by spores. Examples: *Mucor* (pin mould), *Penicillium* (antibiotic-producing mould), yeast.
- Store carbohydrate as glycogen.

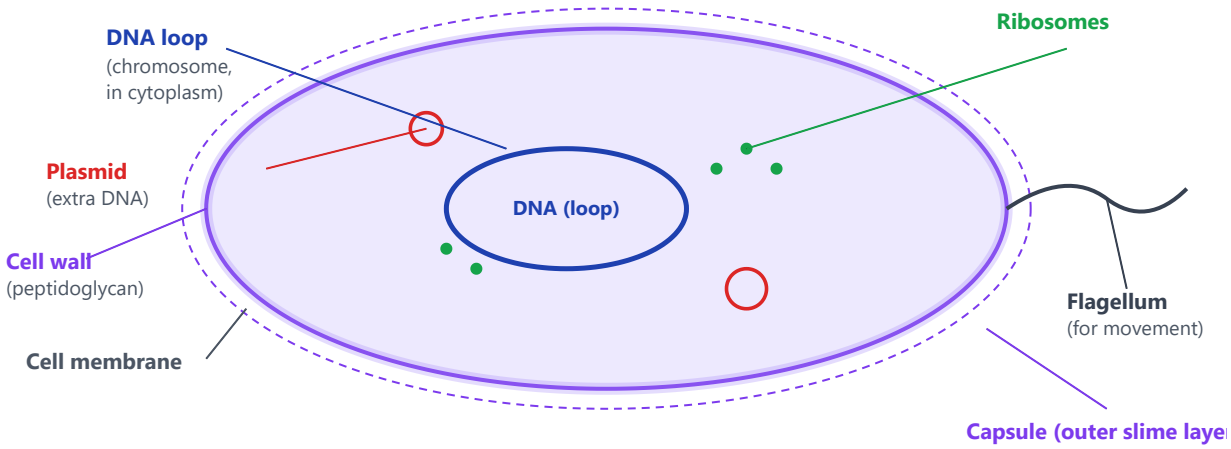
The Protoctista kingdom

- Eukaryotic, usually unicellular organisms. A very varied kingdom; sometimes called a "dumping-ground" for eukaryotes that do not fit in the other kingdoms.
- Some are **plant-like** (contain chloroplasts; photosynthesise) — e.g. *Chlorella*, *Spirogyra*.
- Others are **animal-like** (feed by ingestion) — e.g. *Amoeba*, *Plasmodium* (malarial parasite).

The Prokaryota kingdom (bacteria)

- Prokaryotic: **no true membrane-bound nucleus**. DNA is a single circular loop in the cytoplasm, plus smaller circular **plasmids**.
- No mitochondria, no chloroplasts, no other membrane-bound organelles.
- Cell wall made of **peptidoglycan** (murein), not cellulose or chitin.
- Examples: *Lactobacillus* (used in yoghurt-making), *Pneumococcus* (causes pneumonia), *E. coli*.

Structure of a Bacterial Cell



1.1.3 Pathogens — Disease-Causing Organisms

A **pathogen** is an organism that causes infectious disease in another organism. Pathogens can come from almost every kingdom (bacteria, fungi, protoctists) and also from viruses (which aren't in any kingdom).

Pathogen	Type	Disease	Host
Plasmodium	Protoctist	Malaria	Humans (spread by *Anopheles* mosquitoes)
HIV	Virus	AIDS (suppression of immune system)	Humans
Tobacco mosaic virus (TMV)	Virus	Mosaic discolouration of tobacco leaves	Plants (tobacco, tomato)
Influenza virus	Virus	Flu	Humans, birds
Pneumococcus	Bacterium	Pneumonia	Humans
Vibrio cholerae	Bacterium	Cholera	Humans
Trichophyton	Fungus	Athlete's foot (tinea pedis)	Humans

Why are viruses not considered truly alive?

- A virus is simply **nucleic acid (DNA or RNA) inside a protein coat (capsid)** — sometimes with a lipid envelope. It is not made of cells.
- It has no cytoplasm, no organelles, no ribosomes. It cannot respire, feed, grow or move.
- It can only reproduce **inside a living host cell** by hijacking the host's machinery. So it does not carry out most life processes independently.
- For these reasons most biologists place viruses outside the five kingdoms.

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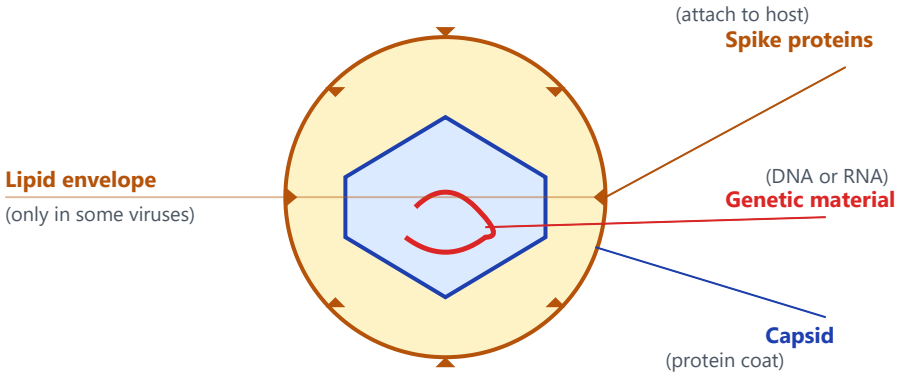
Structure of a Virus (HIV-like)



Case Study: Malaria and *Plasmodium*. *Plasmodium* is a protoctist transmitted to humans by the bite of an infected female *Anopheles* mosquito. It infects liver cells, then red blood cells, causing fever, chills and anaemia. Malaria killed approximately 608 000 people worldwide in 2022, mostly children in sub-Saharan Africa. Control strategies include insecticide-treated bed nets, indoor residual spraying, antimalarial drugs, and — more recently — the RTS,S and R21 vaccines.

Case Study: HIV and AIDS. HIV is a retrovirus that targets helper T-lymphocytes, gradually weakening the immune system. Without treatment, infected people eventually develop AIDS (acquired immune deficiency syndrome), leaving them vulnerable to opportunistic infections. Antiretroviral therapy (ART) suppresses the virus so people live long, healthy lives and greatly reduces transmission to partners and newborns.

1.1.4 Identifying Organisms Using Keys

A **dichotomous key** lets you identify an organism by answering a series of paired yes/no questions, each leading to two choices (hence "dichotomous"). Good keys focus on observable, reliable features, such as cell wall composition, number of legs, leaf shape, or wing pattern.

Example dichotomous key for five kingdoms

1. **Are cells present?**

   Yes → Go to 2. No → *Virus* (not in any kingdom).

2. **Does the organism have a nucleus?**
   Yes → Go to 3. No → *Prokaryota*.
3. **Does the organism have chloroplasts and a cellulose cell wall?**
   Yes → Go to 4. No → Go to 5.
4. **Is the organism multicellular?**
   Yes → *Plantae*. No → *Protoctista* (plant-like).
5. **Does the organism have a chitin cell wall and feed by absorption?**
   Yes → *Fungi*. No → *Animalia* or *Protoctista* (animal-like) depending on whether multicellular or unicellular.

> **Exam Tip:** In the exam, you'll often be shown an unfamiliar organism and asked to classify it. **Read the whole question** — key features (e.g. "has a cell wall of chitin" or "has no nucleus") are planted precisely so you can place the organism in the correct kingdom.

1.1.5 Summary — Why Classification Matters

- **Communication:** Biologists worldwide can refer to the same species unambiguously using the two-part Latin binomial name (genus + species), e.g. *Homo sapiens*, *E. coli*.
- **Evolutionary relationships:** Classification reflects ancestry — organisms in the same kingdom share a more recent common ancestor than those in different kingdoms.
- **Ecology and conservation:** Categorising species allows us to identify and protect ecosystems and endangered species.
- **Medicine:** Knowing whether a disease is caused by a bacterium, virus, protoctist or fungus determines the correct treatment — antibiotics only work against bacteria, not viruses.

> **Quick recall check:** Can you *list MRS GREN*, *state the five kingdoms*, *give one example organism* from each kingdom, and *say why viruses aren't considered alive*? If yes, you have the core of this subtopic. Now move on to the diagnostic.

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